

The supernova history of Cygnus OB2

from Gaia DR3

Progress briefing · WP1-WP6 complete · 2026-07-29

The pitch, in one paragraph

The Cygnus region emits gamma rays up to PeV energies, and the leading explanation (Härer et al. 2025) requires a supernova to have exploded inside Cygnus OB2 within the last $\sim 100\,000$ years. Nobody has checked whether the association can actually supply one. We do it by counting the stars that are missing. Gaia DR3 gives us 2 112 members; we measure each star's extinction and mass, fit the initial mass function where the census is complete ($2\text{--}8\,M_{\odot}$), and extrapolate it upward. Stars heavier than the isochrone turnoff should exist but do not — they have already exploded. Counting them gives the supernova history directly. The critical test is that the same extrapolation must correctly predict the massive stars that are still alive, between $8\,M_{\odot}$ and the turnoff — a genuinely out-of-sample check, because those stars never enter the fit. It now closes to about 10 per cent. The answer that follows is roughly 6 supernovae in the last million years, and a ~ 50 per cent chance that one of them went off inside Härer's 100 000-year window.

Where we are

WP1-WP2	sample and membership	2 112 members, 3 subgroups	done
WP3	per-star extinction	median $A_V = 6.0$ mag	done
WP4	ages and masses	2.5 – 4.0 Myr	done
WP5	IMF normalisation	response-aware, accepted	done
WP6	census closure + runaways	closes to 10%, gate met	done
WP7	the supernova ledger	N_{SN} posterior	next
WP8-WP10	validation, verdict, paper		later

Everything in this document is generated directly from the pipeline's accepted artifacts, so it cannot drift from the code.

1. The idea

Why counting missing stars measures supernovae

A supernova leaves no permanent mark that is easy to find. But the star that exploded is permanently gone — and we can notice its absence.

The logic in four steps

1. Stars form with a known distribution of masses, the initial mass function: many small stars, few large ones, $dN/dM \propto M^{(-\alpha)}$.
2. Massive stars die fast. At a given age, every star above a threshold mass — the isochrone turnoff — has already exploded.
3. We measure the IMF where our census is complete (2–8 $M_{\odot}$) and extrapolate it above the turnoff.
4. That extrapolated number is the number of supernovae.

Why the middle range is the whole game

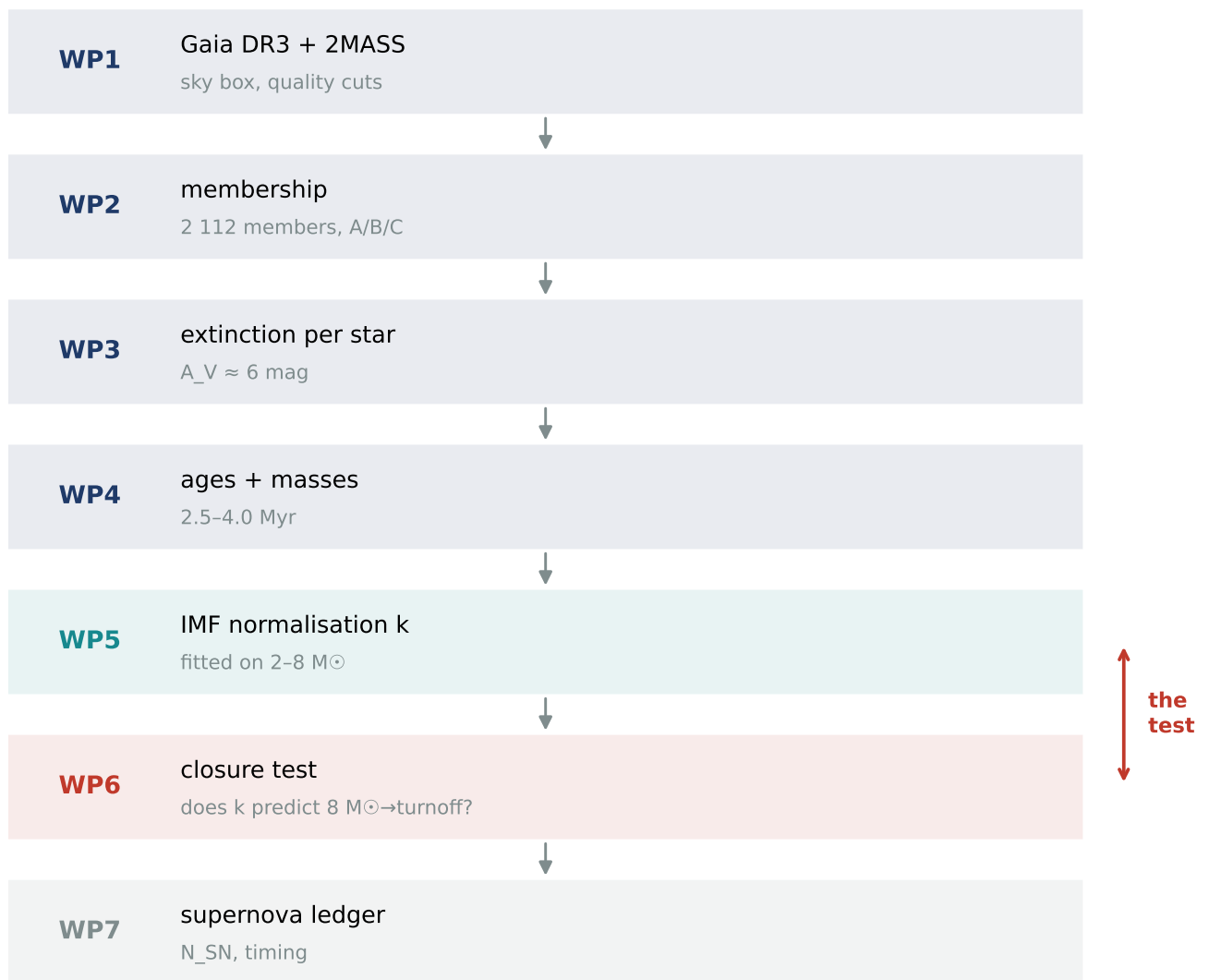
Between 8 $M_{\odot}$ and the turnoff, stars are massive but still alive. The extrapolation must predict how many of those we should see. Those stars never enter the fit, so this is a real out-of-sample test:

if the prediction fails there, the extrapolation above the turnoff cannot be trusted, and the supernova count means nothing.

What makes it hard

- Cygnus OB2 sits behind ~6 magnitudes of dust, and it varies star to star
- the association is three subgroups of different ages, not one cluster
- every cut that removes faint stars also removes low-mass stars, which tilts the very quantity we are measuring
- so completeness cannot be assumed: it must be measured by injecting synthetic stars and seeing how many come back

2. The pipeline

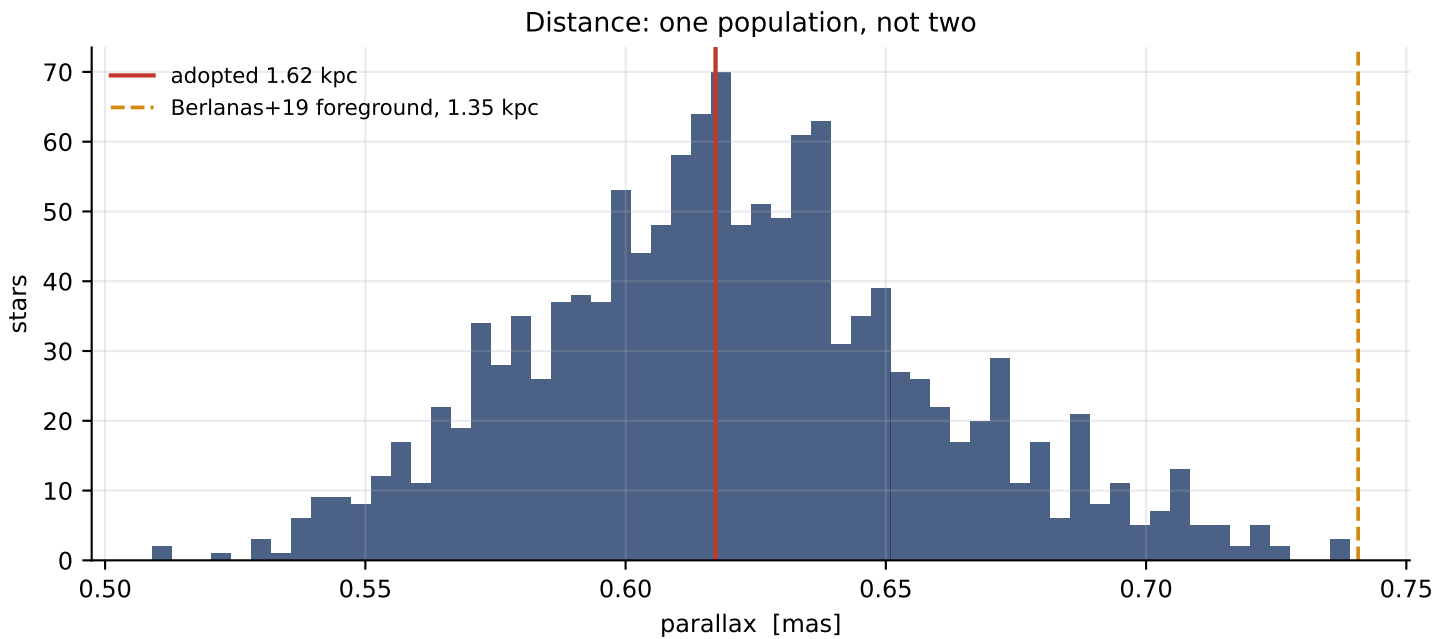
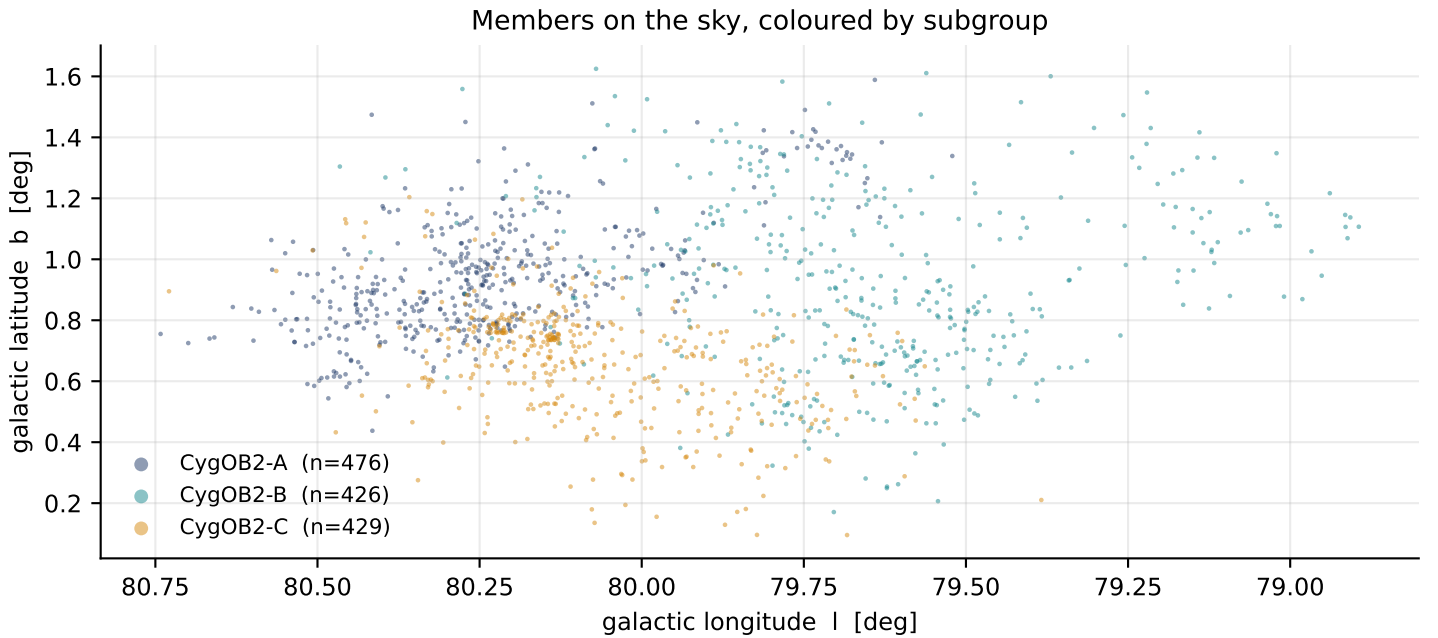


Each stage consumes only accepted outputs of the one above.

WP5 fits the normalisation k using stars from 2 to 8 solar masses, where the census is complete. WP6 then asks whether that same k correctly predicts the massive stars still alive above 8 $M_{\odot}$ — a range that never entered the fit. Only if it does can we extrapolate above the turnoff, which is what WP7 turns into a supernova count.

Every stage is gated: it does not proceed until its own checks pass, and every number carries a provenance record with input checksums.

3. WP1-WP2 — the sample

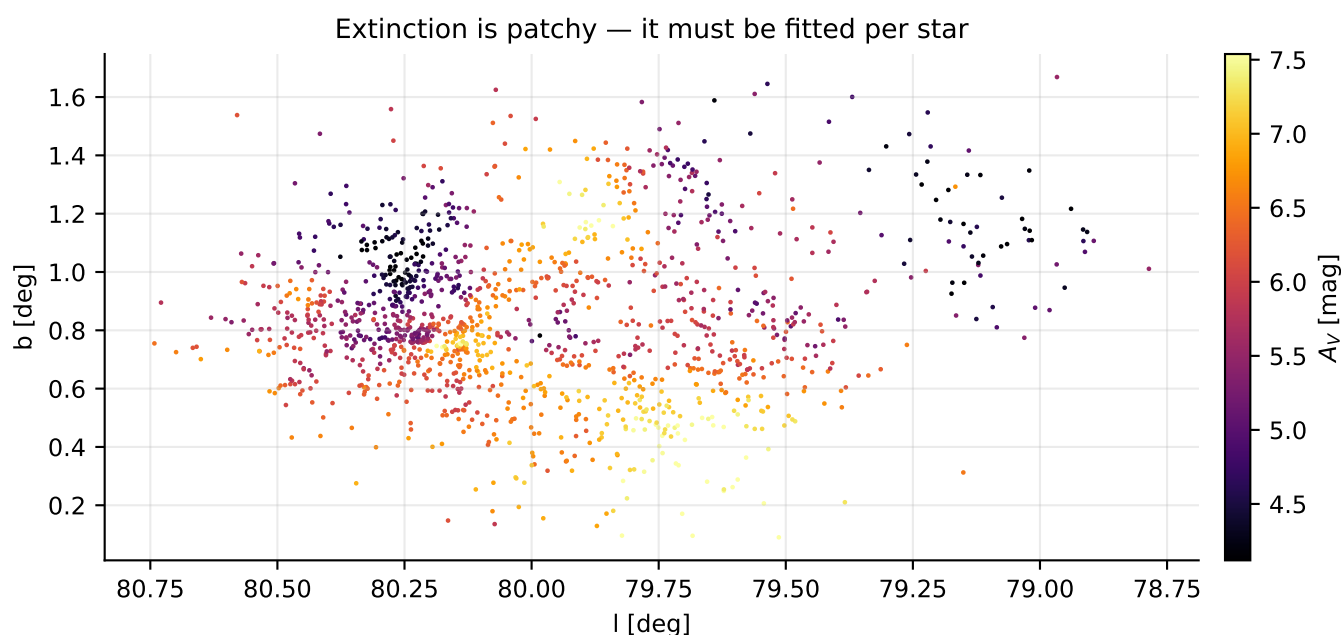
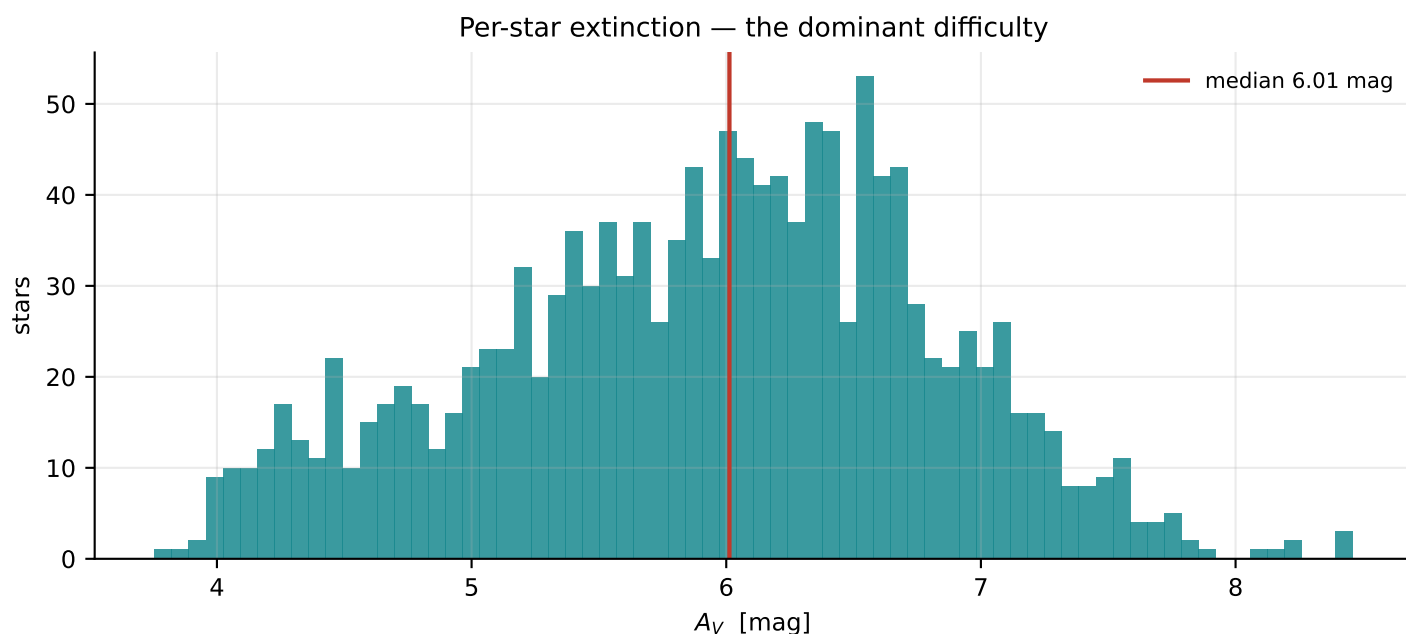


1331 members above 50% membership probability, split into three subgroups by position and proper motion. Membership is a probability, never a hard cut — hard cuts on a magnitude-correlated quantity would bias the mass function.

The lower panel tests a published claim that Cyg OB2 splits into two populations at 1.35 and 1.76 kpc. We find one population at 1.62 kpc with a 45 pc depth; a two-component fit collapses to components 31 pc apart.

Caveat we record: parallax is one of the membership clustering features, so a foreground group would have been removed before this test ran. It shows no split among members — not that none exists.

4. WP3 — extinction

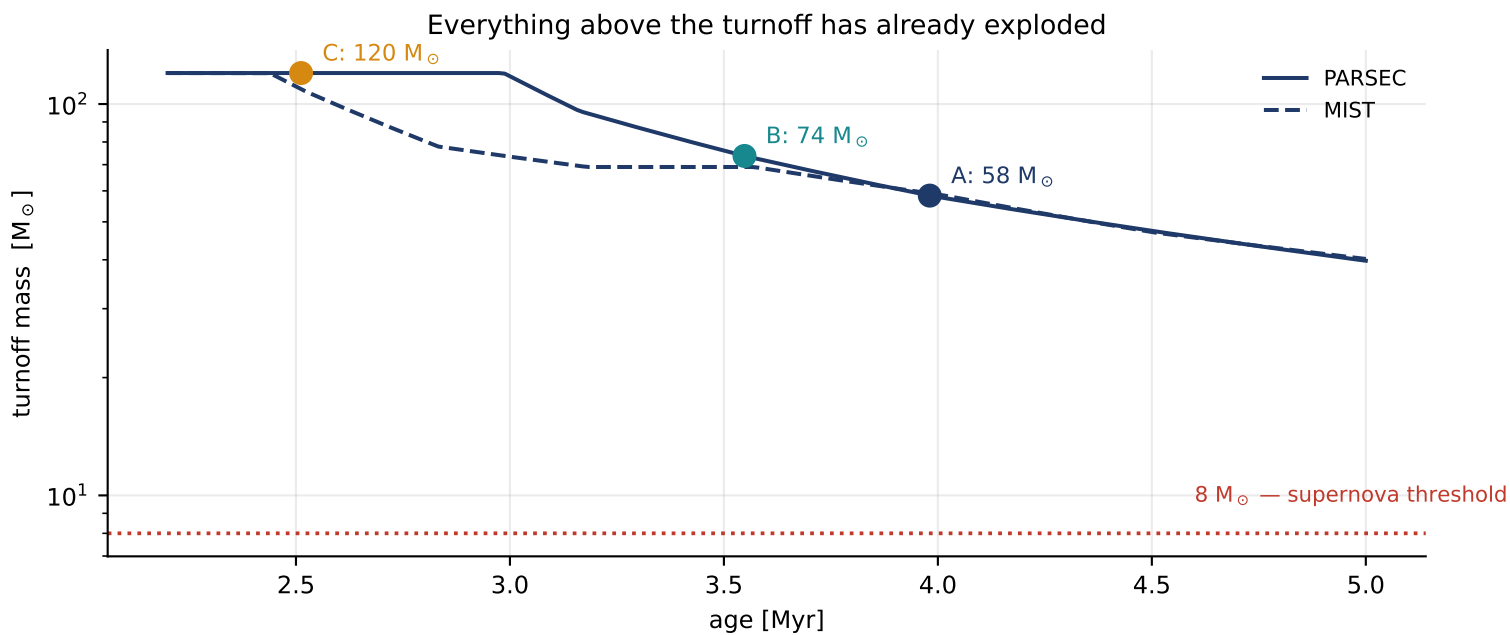
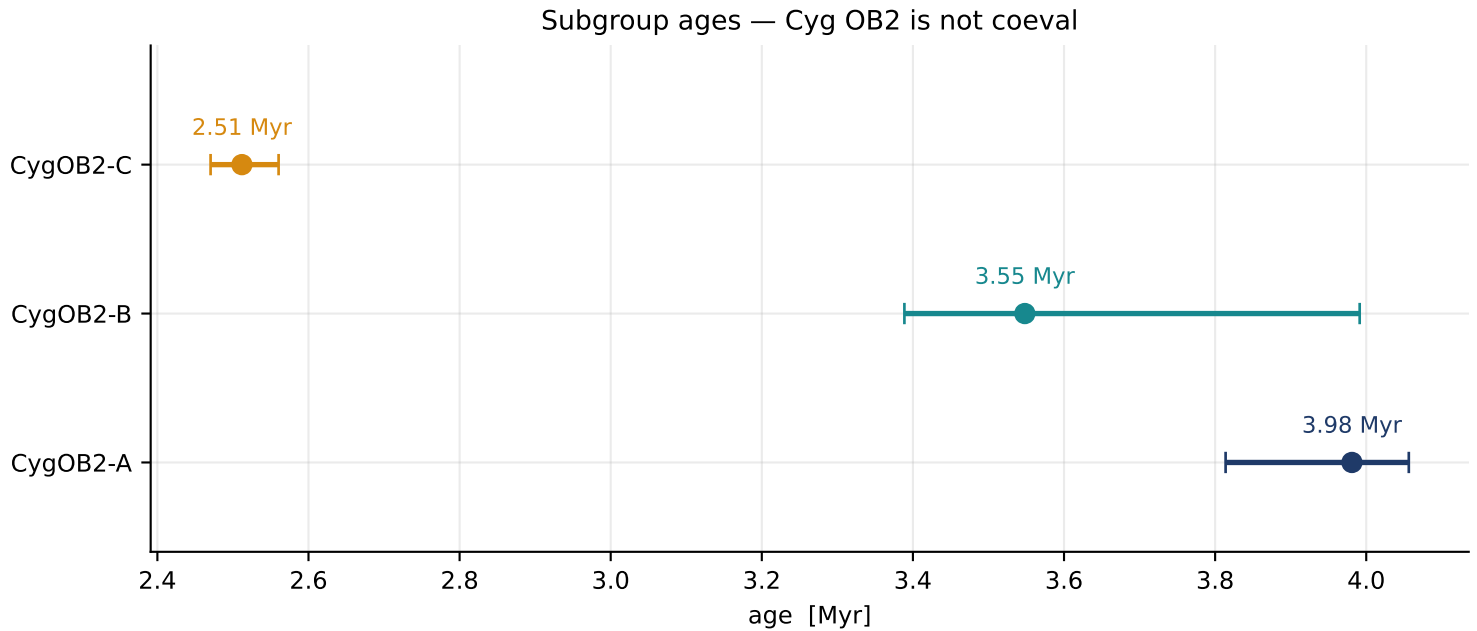


Median $A_V = 6.0$ mag; the 10th–90th percentile spans 4.65–7.03.

That is roughly a factor 250 of dimming in the V band, and it varies across the field, so a single average value would be useless.

Each star gets its own extinction posterior, fitted against a template library with a spatial prior built from spectroscopically classified anchor stars. This stage needed five repair iterations to get right and is the single largest source of systematic uncertainty downstream.

5. WP4 — ages and the turnoff



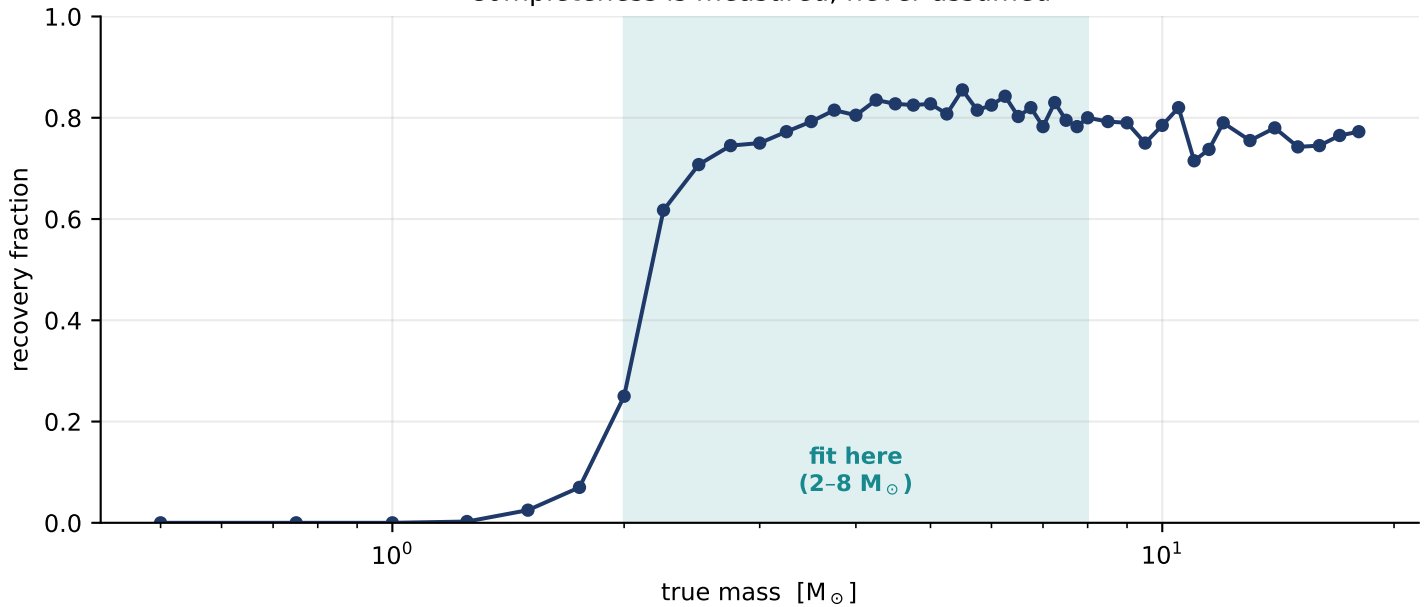
The turnoff is the physics. A star above it has died; below it, it is still shining. So the age of each subgroup sets how many supernovae it has already produced.

CygOB2-C is 2.5 Myr old and its turnoff is still above the $120 M_{\odot}$ upper limit of the IMF — nothing in C has exploded yet. Essentially all the supernovae come from A and B.

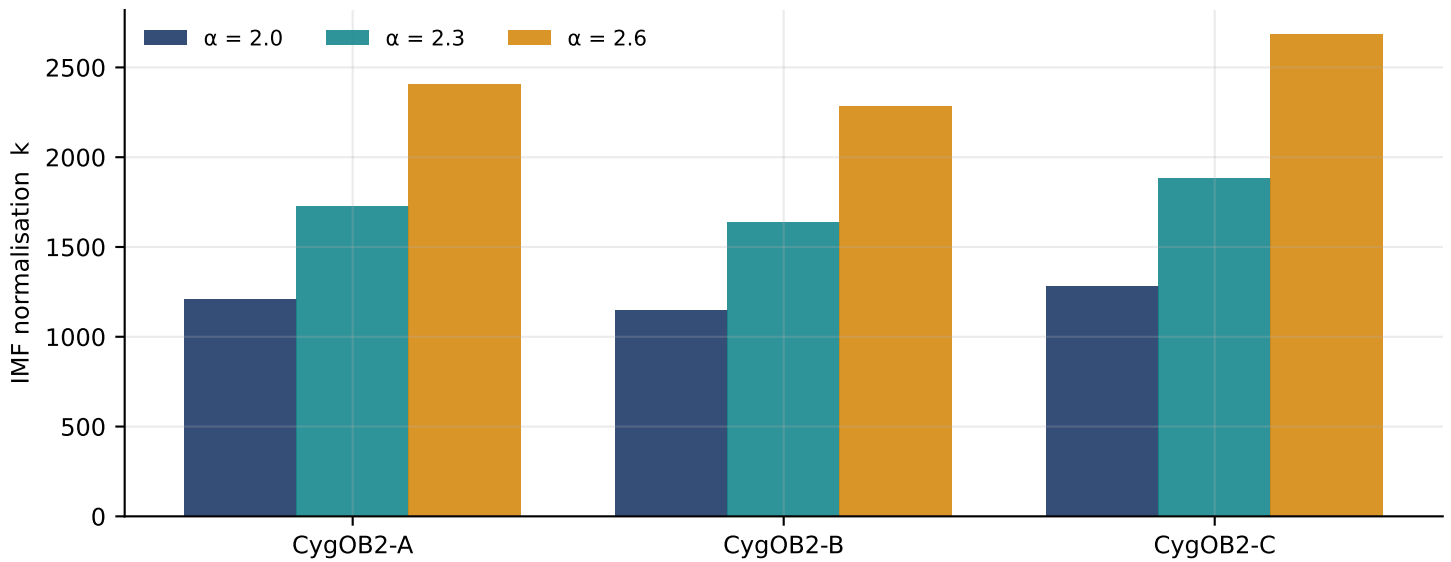
*The two isochrone families disagree by up to 30% at these ages.
We carry both in parallel to the end and never average them.*

6. WP5 — the IMF normalisation

Completeness is measured, never assumed



The fitted normalisation, per branch



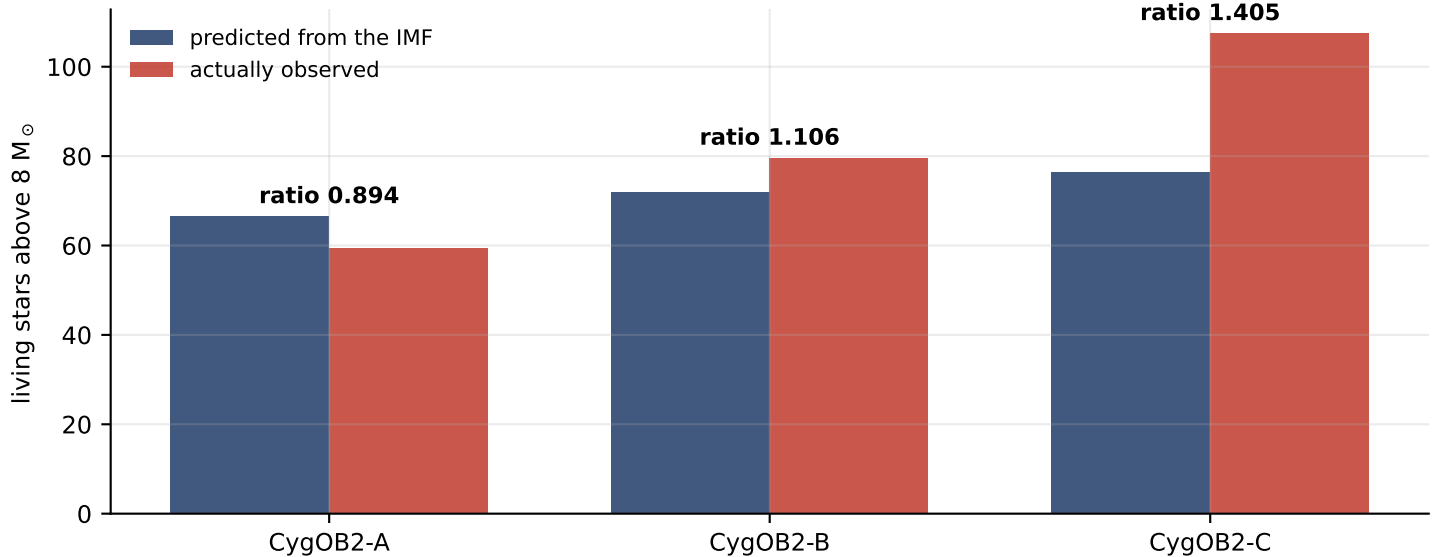
We inject synthetic stars of known mass and count how many survive the entire pipeline. That gives the response — the probability that a star of true mass M is recovered, and what mass it is then assigned.

The normalisation k is fitted only in 2-8 $M_{\odot}$ (shaded), through that response, with a Poisson likelihood. Dividing an observed count by an average completeness is forbidden in this project: the response also scatters masses across bin edges, and in 6 of 54 cases that correction would have the wrong sign.

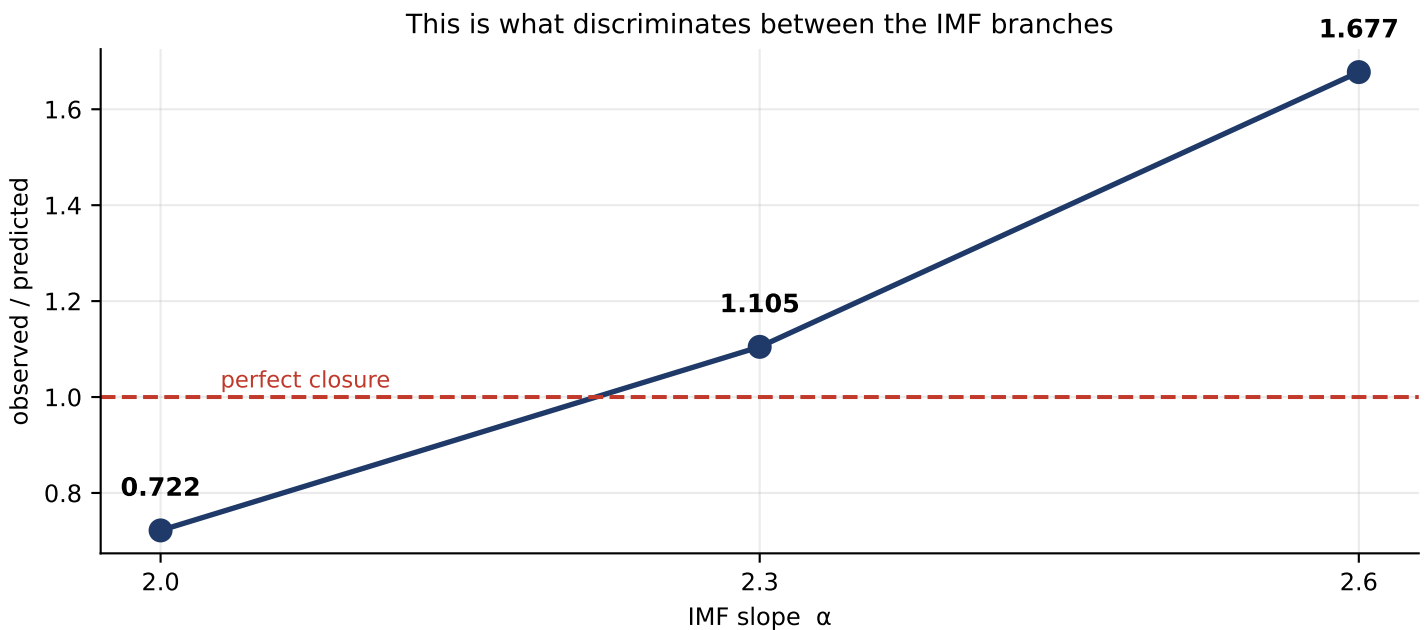
The slope α is not fitted. It is carried as three parallel branches.

7. WP6 — the closure test

The out-of-sample test: does the extrapolation work?



This is what discriminates between the IMF branches



The load-bearing result. The IMF fitted on 2-8 M_⊙ is extrapolated into 8 M_⊙ → turnoff and compared with what we actually see there. Those stars never entered the fit.

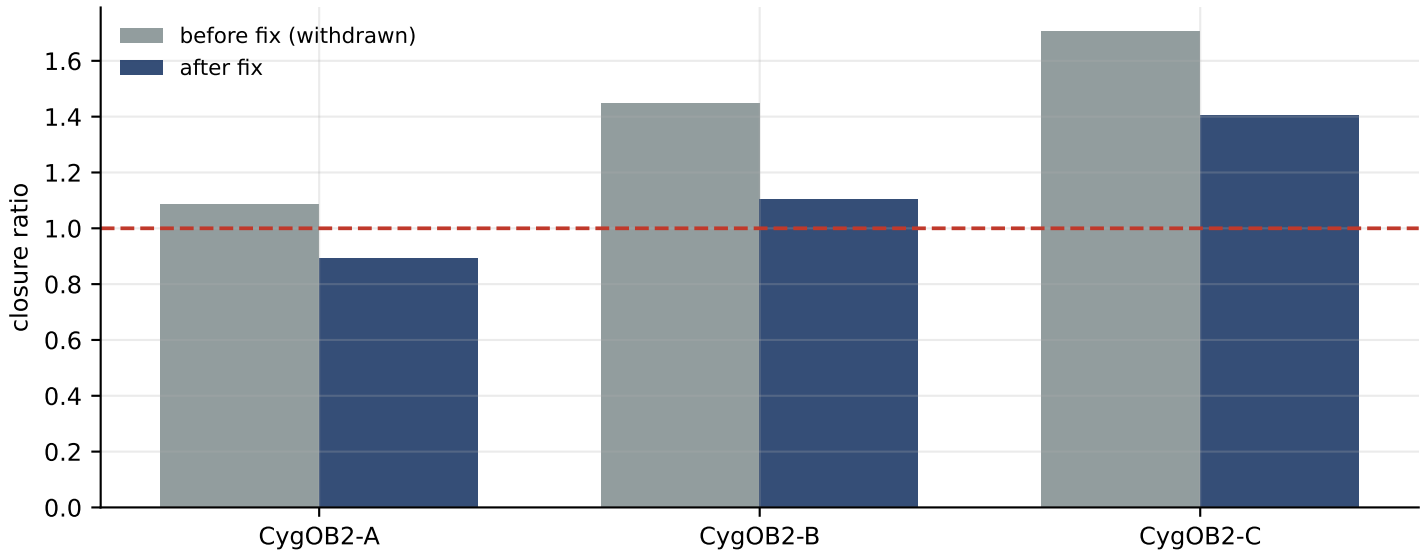
The census closes to about 10% (grid median 1.105), and the closing slope is $\alpha \approx 2.23$ — close to the standard Salpeter value of 2.35.

The lower panel is arguably the most useful thing WP6 produces: $\alpha = 2.6$ over-predicts badly and $\alpha = 2.0$ under-predicts, so the census itself selects between branches that the fit alone could not.

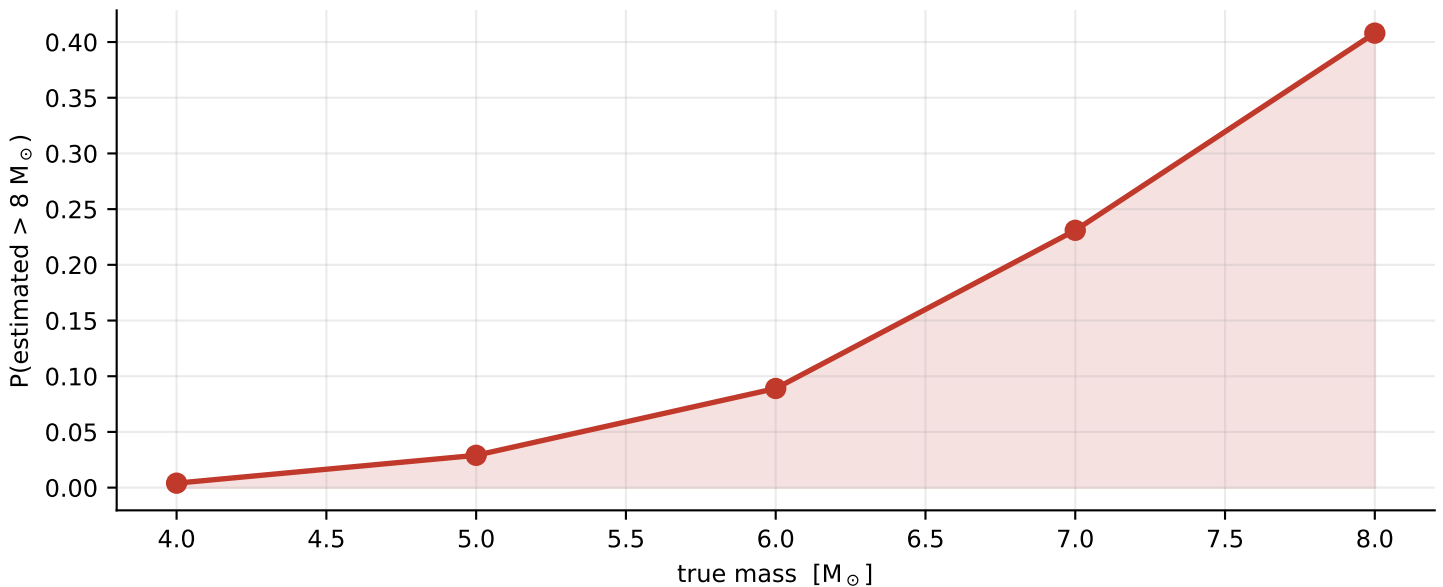
CygOB2-C remains at 1.405 while A sits at 0.894 — they disagree in direction, so no single mechanism explains both. This is the main open scientific question.

8. Two bugs we found and fixed

Bug #17: the integral truncated at the threshold



Stars below $8 M_{\odot}$ that get measured above it



These were found by our own checks, and both changed the answer.

Bug #17 — the closure integral started at $8 M_{\odot}$, but the observed side

counted every star measured above $8 M_{\odot}$ whatever its true mass. A $7 M_{\odot}$ star scattered upward was counted and never predicted (lower panel).

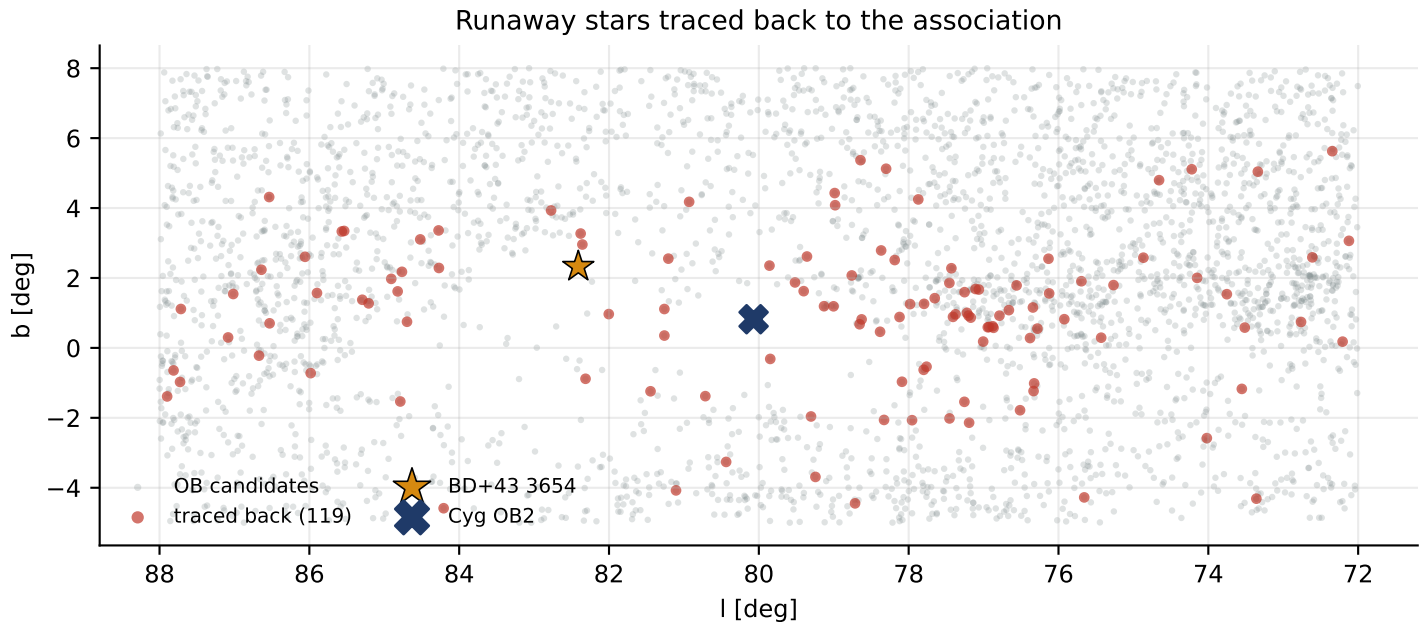
Fixing it moved the grid median from 1.444 to 1.105 and the closing slope from 2.07 to 2.23 — about three quarters of what looked like a 'shallower-than-Salpeter IMF' was this bug.

Bug #16 — the runaway search used absolute proper motions, but Cyg OB2's

own bulk motion is larger than a typical ejection signature. It was caught by an external check (next page).

Both results were formally withdrawn and the superseded numbers are kept on record, not deleted. Three published values are affected.

9. Runaways — and the check that caught a bug



BD+43 3654 — the external check that caught bug #16

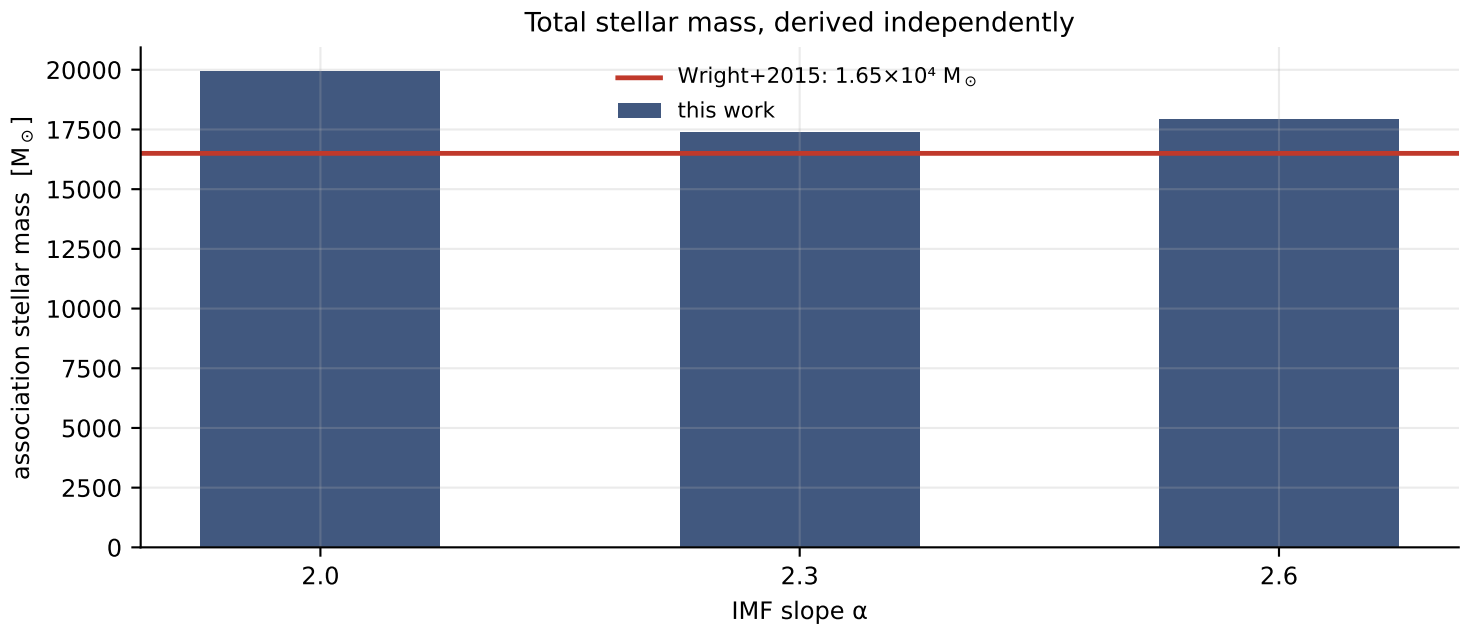
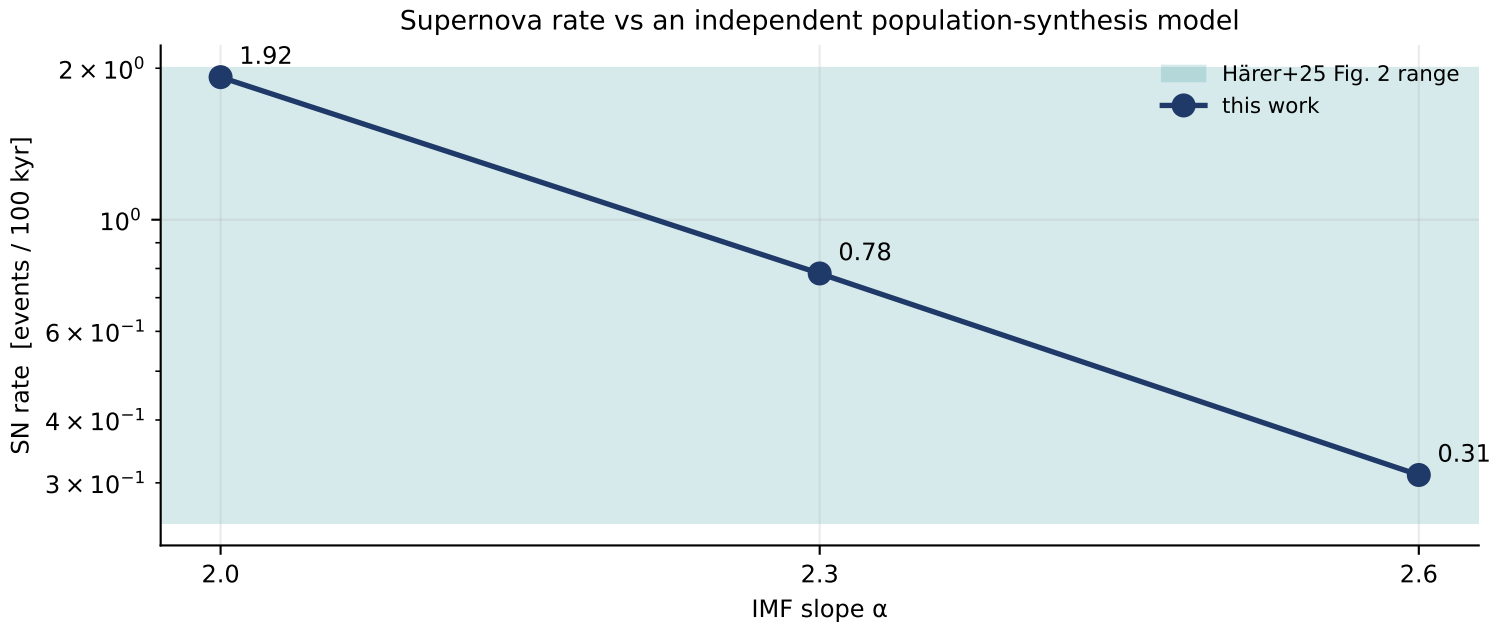
	measured here	literature
recovery probability	1.000	—
ejection velocity	38.8 km/s	~40 km/s
flight time	1.36 Myr	1.6 Myr
position match	0.04 arcsec	—

Some massive stars were ejected from the association and are still alive elsewhere. They must not be counted as dead. We trace proper motions backwards and recover 119 candidates; after subtracting a chance-alignment rate measured from control fields, 54.9 are genuine.

BD+43 3654 is the textbook Cyg OB2 runaway, with an independently known velocity and flight time. Under the buggy version it scored exactly **0.000**. **After the fix it scores 1.000 and reproduces both published** numbers.

This is why external checks are not optional: every internal diagnostic passed while the result was inverted.

10. Independent cross-checks



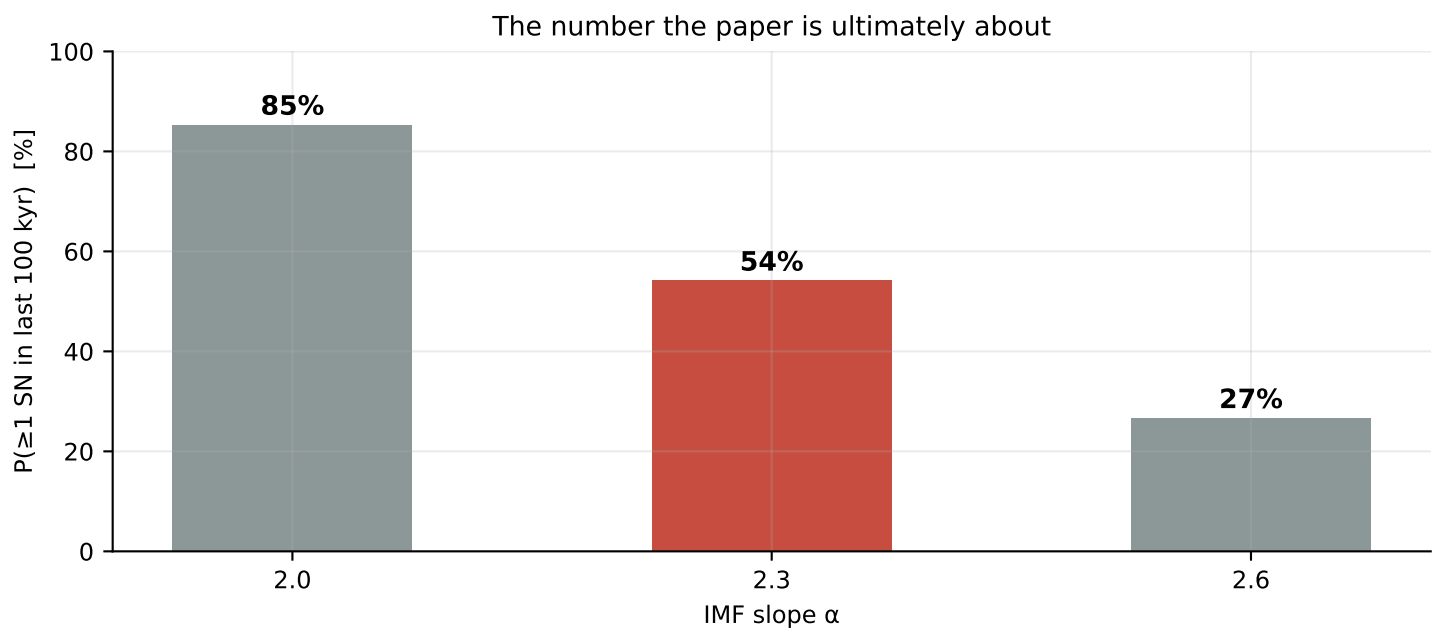
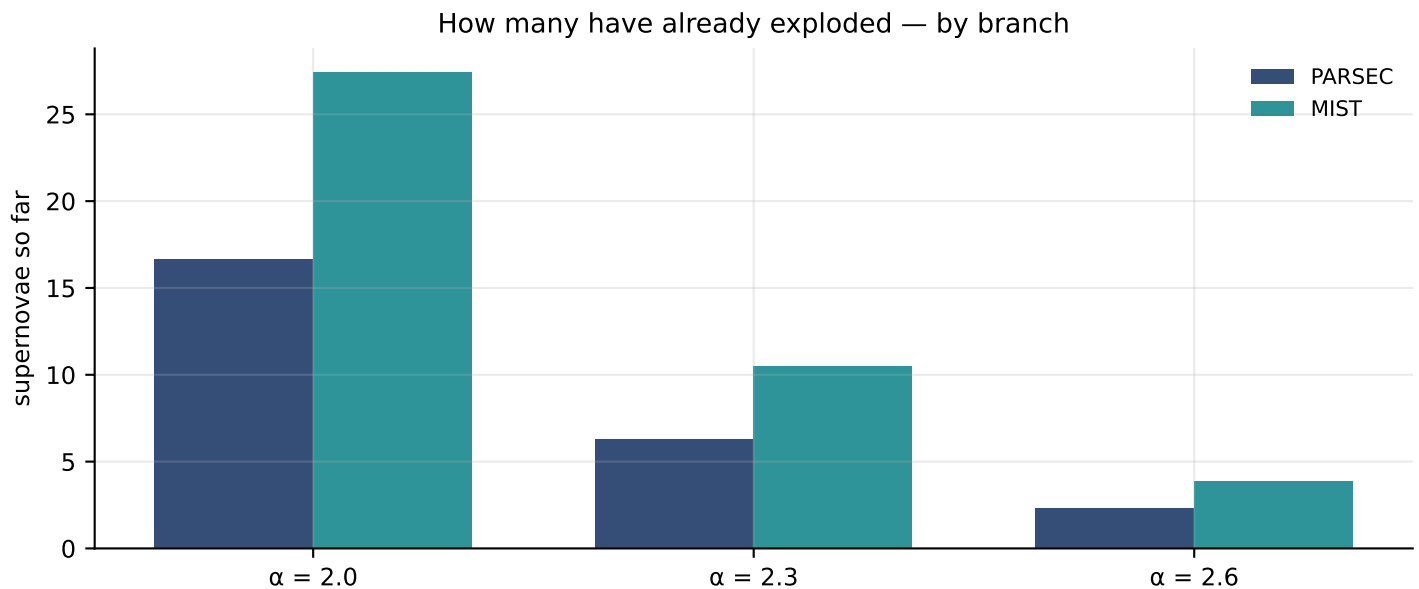
Three published results, none of which we tuned to, and all of which agree.

- Supernova rate — Härer+25 compute it from BPASS population synthesis for an assumed cluster mass. All three of our branches land inside their plotted range; our baseline gives 7.8 SNe/Myr against their stated 'several per Myr'.
- Association mass — $1.74 \times 10^4 M_{\odot}$ from our fit, against the 1.65×10^4 they assumed. A 5% agreement by a completely different route.
- Systemic proper motion — our -2.7067 mas/yr against Orellana+21's -2.71 ± 0.02 from Gaia DR2: agreement to 0.003 mas/yr.

The rate and the mass depend on the fitted k in different ways, so a broken extrapolation would generally break them inconsistently.

*Distance is the one mild tension: we get 1.62 kpc, Orellana 1.67 kpc.
Traced to the DR2 vs DR3 parallax zero point; carried as a 2.4% systematic rather than adjusted.*

11. Where this is heading (WP7 preview)



These are illustrative estimates from the fitted normalisation, not yet the WP7 result — WP7 will do this by discrete stochastic sampling, which matters at these small numbers.

Roughly 6 supernovae at the baseline branch, essentially all from subgroups A and B in the last ~ 1 Myr; C has produced none.

The lower panel is the paper's target statement: the probability that at least one went off inside Häner's 100 000-year window. At the baseline it is about 50% — genuinely marginal, which is why the age sensitivity will have to be shown explicitly rather than marginalised away.

Note the branch spread: N_{SN} ranges over a factor of 12 across the carried branches. That, not statistical noise, dominates the answer.

12. Status, honesty, and what is next

An honest account of what holds and what does not

What is solid

- the sample, the extinctions, the ages and the normalisation are all accepted through formal gates, with checksummed provenance
- the out-of-sample closure test works at the $\sim 10\%$ level
- three independent literature cross-checks agree: supernova rate, association mass, and systemic proper motion
- the runaway search reproduces the textbook Cyg OB2 runaway

What is still open

- CygOB2-C sits at 1.405 while A sits at 0.894. They disagree in direction, so no single mechanism explains both. This is the main scientific question left. Candidates: subgroup labelling, foreground contamination in distance, and the 2.4% distance systematic.
- a chain re-run (repair_v7) is under way: a pre-registered test showed the binary fraction below $8 M_{\odot}$ shifts the threshold-crossing response by 9.9%. It moves the closure ratio to ~ 1.074 and the supernova count by only 0.5%.

How we work, and why it is worth mentioning

- every test is pre-registered: predictions and thresholds are written to a provenance file before the code runs, so a result cannot be quietly reinterpreted afterwards
- when a prediction fails, it is recorded as failed. Two have been.
- superseded numbers are formally withdrawn and kept on record
- three separate bugs were caught this way, two of which changed the headline answer

Next steps

1. WP7 — the supernova ledger: N_{SN} posterior, explosion timeline, and the probability of one within the last 100 kyr, per branch
2. WP8 — validation against the pulsar PSR J2032+4127, the γ -Cygni remnant, and the ^{26}Al measurements
3. WP9-WP10 — the verdict and the manuscript